QMAS

Quality Metrics & Analysis System

Operational Run Sheet | Library: RWYNN01 | IBM i V7R5

1. System Overview

QMAS loads manufacturing defect data from a CSV file into DB2 for i and provides a 5250 green-screen interface for shop floor managers to search, filter, and drill into defect records.

Component	What It Does
CLWRAP	CL wrapper — sets up the environment and submits QMASBATCH
QMASBATCH	Batch program — reads CSV data, calculates costs, loads DEFECTPF
QMASTCOST	Subprogram — calculates total cost per defect using severity and stage multipliers
QMASINT	Interactive program — 5250 screen for filtering and viewing defect records
DEFECTPF	DB2 table in library RWYNN01 — stores all loaded defect records

2. Running the Batch Job

The batch job reads the defect CSV file, validates each record, calculates the total cost, and loads the data into DEFECTPF. Run it using CLWRAP which handles all library setup automatically.

2.1 Submit to Batch (Recommended)

Use SBMJOB to submit the job to the batch queue. This runs the job in the background and returns you to the command line immediately:

```
SBMJOB CMD(CALL PGM(RWYNN01/CLWRAP))
JOB(BATCHRUN1)
JOBD(QGPL/QDFTJOBD)
OUTQ(QPRINT)
```

2.2 Schedule an Automatic Nightly Run

To schedule QMASBATCH to run automatically at a set time each day or overnight, use the IBM i Job Scheduler (ADDJOBSCDE):

```
ADDJOBSCDE JOB(BATCHRUN1)
```

```
CMD(CALL PGM(RWYNN01/CLWRAP))
FRQ(*WEEKLY)
SCDDAY(*MON *TUE *WED *THU *FRI)
SCDTIME(020000)
```

*Note: SCDTIME(020000) = 2:00 AM. Change to suit your schedule. FRQ(*DAILY) runs every day. FRQ(*WEEKLY) with SCDDAY controls which days of the week.*

To view or manage scheduled jobs:

```
WRKJOBSCDE
```

2.3 Monitor the Job

After submitting, check the job status:

```
WRKACTJOB JOB(BATCHRUN1)
```

2.4 Review the Control Report

When the job completes a control report is written to the spool file showing totals for every run:

```
WRKSPLF
```

Take option 5 next to the RPTOUT entry to view the report. It shows:

Column	Description
SEQ	Record sequence number
DEFID	Defect ID from the CSV
STATUS	LOADED or REJECTED
MESSAGE	Blank for loaded records, rejection reason for rejected records

Summary totals at the bottom of the report:

Total	Description
TOTAL RECORDS PROCESSED	Every record read from the CSV including the header skip
TOTAL RECORDS LOADED	Records successfully inserted or updated in DEFECTPF
TOTAL RECORDS REJECTED	Records that failed validation or SQL — see MESSAGE column
TOTAL FINANCIAL VALUE	Sum of all TCOST values for loaded records

3. Using the Interactive Program (QMASINT)

QMASINT is a 5250 green-screen program that lets shop floor managers search, filter, and drill into defect records loaded by QMASBATCH.

3.1 Launch

Call QMASINT from any 5250 command line. Make sure QMASBATCH has been run first so DEFECTPF has data:

```
CALL PGM(RWYNN01/QMASINT)
```

3.2 Screen 1 — Filter Menu

Enter one or more filter values and press Enter to search. Leave any field blank to return all records.

Field	Valid Values
Production Line	Enter a line code e.g. LN01, LN02. Leave blank for all lines.
Severity	1 = Critical, 2 = Major, 3 = Minor. Leave blank for all.
Detection Stage	RW = Raw Material, IP = In Process, FG = Finished Goods. Leave blank for all.

3.3 Screen 2 — Defect List

A scrollable subfile showing all defects matching your filters. Type 5 next to any row and press Enter to view the full detail record.

3.4 Screen 3 — Detail View

Displays the complete defect record including all costs. A CRITICAL ESCAPE warning appears in reverse image when Severity = 1 (Critical) and Stage = FG (Finished Goods) — indicating a critical defect that escaped to the finished goods stage.

3.5 Function Keys

Key	Action
F3	Exit — closes the program from any screen
F5	Refresh — reloads the subfile with the current filters
F12	Go Back — returns to the previous screen
5 + Enter	View Detail — type 5 next to any subfile row to drill down

4. Viewing the DEFECTPF Table Directly

DEFECTPF is the DB2 for i table that stores all loaded defect records. You can view it in two ways:

4.1 Schema Browser in VS Code (IBM i Extension)

In the VS Code IBM i extension, open the Object Browser and navigate to:

Location	Path
Library	RWYNN01
Section	Tables
Table	DEFECTPF

Note: In the IBM i extension this appears under: RWYNN01 > Tables > DEFECTPF. Right-click the table and select 'View table contents' to browse records directly.

4.2 SQL Query

Run this SQL statement in the IBM i Run SQL Scripts window or STRSQL to view all records in DEFECTPF:

```
SELECT * FROM RWYNN01.DEFECTPF as a
```

Useful variations:

```
SELECT COUNT(*) FROM RWYNN01.DEFECTPF as a
SELECT * FROM RWYNN01.DEFECTPF as a ORDER BY DEFID
SELECT * FROM RWYNN01.DEFECTPF as a WHERE SEV = 1
SELECT * FROM RWYNN01.DEFECTPF as a WHERE STAGE = 'FG' AND SEV = 1
```

Note: The last query returns all Critical Escape records — SEV=1 defects detected at the Finished Goods stage.

4.3 DEFECTPF Column Reference

Column	Type	Description
DEFID	CHAR(8)	Defect ID — primary key
LINE	CHAR(4)	Production line code e.g. LN01
PARTNO	CHAR(10)	Part number
TYPECD	DECIMAL(2,0)	Defect type code
TYPEDESC	CHAR(30)	Defect type description e.g. Surface Contamination

Column	Type	Description
SEV	DECIMAL(1,0)	Severity: 1=Critical, 2=Major, 3=Minor
BCOST	DECIMAL(9,2)	Base cost of the defect
STAGE	CHAR(2)	Detection stage: RW, IP, or FG
RWK	CHAR(1)	Rework flag: Y or N
TCOST	DECIMAL(11,2)	Total cost = BCOST x Severity multiplier x Stage multiplier

5. Quick Reference

Task	Command
Submit batch job	SBMJOB CMD(CALL PGM(RWYNN01/CLWRAP)) JOB(BATCHRUN1)
Check job status	WRKACTJOB JOB(BATCHRUN1)
View control report	WRKSPLF then option 5 on RPTOUT entry
Launch interactive program	CALL PGM(RWYNN01/QMASINT)
View DEFECTPF in SQL	SELECT * FROM RWYNN01.DEFECTPF as a
View scheduled jobs	WRKJOBSCDE
Add library to list	ADDLIBLE LIB(RWYNN01)